

Why the Database is Critical to CricGeek

Architectural Role, System Dependencies, & Product Segment Implementation

Core Platform Overview

CricGeek is not a basic cricket news site that only displays live scores; it is a **gamified social publishing and AI platform**.

The database serves as the single central memory for the entire application. It connects real-time AI content evaluation, user progression, automated fact-checking, and interactive community features into a unified system.

Impact of Database Disconnection

Without a persistent database layer, the platform loses its core functionality across four vital areas:

- **Published Content Vanishes:** Users cannot save drafts, publish articles, or keep track of their published work.
- **AI Scores Cannot Be Saved:** The AI evaluates article quality, but without persistence, scores disappear—meaning articles cannot be approved or promoted to the public feed.
- **No User Progression:** Writers cannot gain levels, earn badges, or climb platform leaderboards.
- **Fallback to Static Demo Data:** When disconnected, the system quietly falls back to displaying hardcoded "demo" data instead of real user contributions.

Database Implementation by Product Segment

The CricGeek database partitions its architecture into **31 specific tables** distributed across 7 core product areas.

Segment	Primary Data Stored	Architectural Purpose
1. User Identity & Security	User accounts, credentials, role assignments	Enforces system-wide RBAC (User, Writer, Admin) permissions
2. Content & Expression	Articles, slugs, match IDs (<code>matchTag</code>), comments, likes	Connects posts with live match data for stat verification
3. AI Quality Engine	13 quality scores, quality bands, claim audit logs	Acts as the gateway trigger (<code>processingStatus === "completed"</code>) for public feed visibility
4. Writer Identity & DNA	XP, levels, badges, Writer DNA profiles	Customizes AI scoring rules based on author archetype (e.g., Analyst vs. Storyteller)
5. Contests & Leaderboards	Admin competitions, user entries, challenge scores	Dynamically updates writer standings and challenge rankings
6. Live Voice Commentary	Audio streams, speaker lists, transcripts, timing data	Converts spoken audio into text entries and corrects player name pronunciations
7. Cricket Warehouse	Archived scorecards, player statistics	Serves as the ground truth source for automated statement verification

Detailed Segment Breakdown

1. User Identity & Security (Auth)

- **What it stores:** User accounts, hashed passwords, and assigned user roles (`user`, `writer`, `admin`).
- **Why it's needed:** It controls permissions across the platform. For example, only accounts marked as `writer` can publish articles, and only accounts marked as `admin` can create contests or moderate content.

2. Content & Expression (Blog System)

- **What it stores:** Article titles, body text, unique URL slugs, attached match IDs (`matchTag`), user comments, cricket-ball reactions, and saved posts.
- **Why it's needed:** Linking an article to a specific match ID in the database allows the AI engine to pull the official match scorecard and verify whether the statistics written in the article are accurate.

3. AI Quality Engine (EQS & BlogScore)

- **What it stores:** The AI's 13 quality scores (clarity, reasoning, toxicity, stat accuracy), quality bands, and complete audit trails (claim checks and plagiarism scan results).
- **Why it's needed:** The public blog feed strictly filters for posts where `processingStatus === "completed"` in the database. Storing this score is the exact trigger that allows a draft to become visible to the public.

4. Writer Identity & Gamification (Profiles & DNA)

- **What it stores:** Writer levels, experience points (XP), badges, average quality scores, and "Writer DNA" metrics (Analyst, Storyteller, Fan, Debater).
- **Why it's needed:** The system customizes its AI evaluation based on the writer's style stored in the database. An *Analyst* is judged heavily on statistical accuracy, while a *Storyteller* is judged on narrative flow.

5. Contests & Leaderboards

- **What it stores:** Writing competitions created by admins, user submissions, final challenge scores, and prize rankings.
- **Why it's needed:** It dynamically updates writer standings on the public leaderboard as soon as new articles are scored by the pipeline.

6. Live Voice Commentary

- **What it stores:** Live audio commentary sessions, speaker lists, transcribed text, and audio timing data.
- **Why it's needed:** It converts live spoken commentary into permanent text entries and pulls team squad lists from the database to correct mispronounced player names.

7. Historical Cricket Warehouse

- **What it stores:** Archived match scorecards and historical player statistics.
- **Why it's needed:** It serves as the official truth source. When a writer claims *"Bumrah took 5 wickets in 2023,"* the AI checks this table to confirm whether the statement is true or false.